DEPARTMENT OF INFORMATION SYSTEMS

SCHOOL OF COMPUTING - FUTA

B.TECH - SECOND SEMESTER EXAMINATION - 2023/2024 SESSION

IFS510: XML DESIGN PARADIGMS

Instruction:** Read carefully, answer any 3* | ***Time:** 2hrs* | ***2 Units

QUESTION 1 (20Marks)

1a. Explain briefly Document Type Definition (DTD)? And give four ways DTD is different from XML.

1b What are the roles and advantages of XML?

1c. How can both Internal and External DTDs be used in an XML File? Show with an Example.

1d. List 10 XPaths Axes and their functions.

QUESTION 2 (20Marks)

Using the XML code presented in this question

```
<body type="anthology">
        <div type= "poem"><head>OLD ANTHEM</head>
                <lg type="stanza">
                        <l n="1">O God of Creation.</l>
                        <l n="2">Na chop I wan chop</l>
                        <l n="3">That flies in the night</l>
                        <l n="4">Make Human no Quench</l>
                </lg>
                <lg type="stanza">
                        <l n="5">Writing 9 Course in a week</l>
                        <l n="6">I've chosen the life of a Chosen</l>
                        <l n="7">And his dark secret love</l>
                        <l n="8">Does thy CGPA destroy.</l>
                </lg>
        </div>
        div type= "shortpoem"><head>NEW ANTHEM</head>
                <lg type="couplet">
                        <l n="1">Nigeria we Hail Thy</l>
                        <l n="2">The invisible Season of Survival</l>
                </lg>
        </div>
</body>
```

2a. Draw the detailed XML Structure

2b. write the expression that (i) will return the value of ***n*** greater than 5, (ii) will select all the attributes ***type*** (iii) will select all the attribute ***n*** with ***n=2*** **(iv)** will select all the descendant with ***type*** stanza (v) will select the ancestor of the type ***shortpoem***

QUESTION 3 (20Marks)

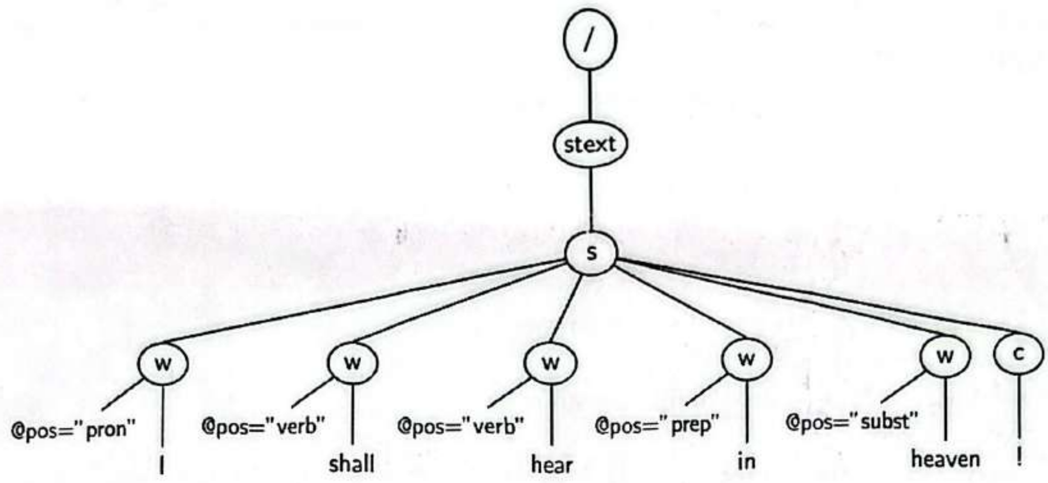
The tree below shows an XML document, drawn according to the XPath data model. It represents a passage of spoken text annotated according to (a simplification of) the mark-up scheme of a National Corpus. This annotates text by grouping it into sentences, indicating individual words and punctuation, along with \part of speech" (pos) information | whether a word is a verb, a substantive (a noun), preposition, and so forth. *NOTE: [subst = substantive; prep = preposition; pos=part of speech; pron=pronoun]*

3a. Write out the tree as an XML document

3b. Write XPath expressions to return the following lists of text strings for the tree above

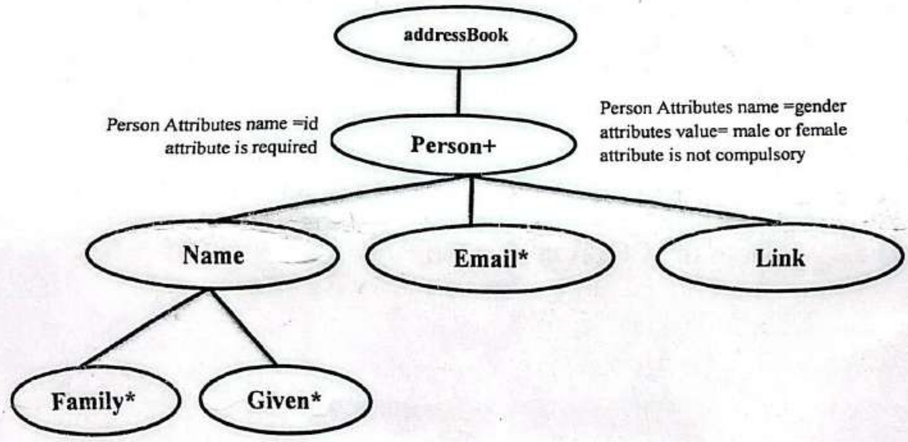
i. All punctuation marks.

ii. All substantives. The pos tag for substantives is \SUBST".
iii. All words appearing in sentences that contain an exclamation mark \!".
iv. All substantives appearing in sentences that contain an exclamation mark \!".
v. All punctuation marks of sentences that contain the word \face".



QUESTION 4 (20 Marks)

4. Create an internal DTD and a sample XML for the TREE BELOW



QUESTION 5 (20Marks)- Using the XML provided in the box

5a. Write a Document Type Definition (DTD) for the XML that is given below

5b. Write an XPath expression that finds Choice elements where the choice number is 1 and the score is greater than 80; where the Choice number is less than 3 and score greater than 70.

```
<Qus>
        <Applicants>
                <Applicant name="Toluojo" appNum="a1" />
                <Applicant name="Ajayiseun" appNum="a2" />
                <Applicant name="SmithiTope" appNum="a3" />
        </Applicants>
        <Choices>
                <Choice applicant="a1" code="IFS102" choiceNum="1" score="80" />
                <Choice applicant="a1" code="IFS101" choiceNum="2" score ="98" />
                <Choice applicant="a1" code="IFS107" choiceNum="3" score ="90" />
                <Choice applicant="a2" code="IFS203" choiceNum="1" score ="70" />
                <Choice applicant="a3" code="IF302" choiceNum="1" score ="85" />
                <Choice applicant="a3" code="IFS304" choiceNum="2" score ="85" />
        </Choices>
</Qus>
```

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF INFORMATION SYSTEMS

B. Tech. Degree Examination; Second Semester 2023/2024 Session
Course Title: Grid & Cloud Computing
Code: IFS 516; **Units:** 2
Instruction: Answer *three* Questions in ALL. **Time:** 2hrs

1a. The department of information systems, FUTA, needs to process large datasets across multiple geographical locations. Explain how Grid Computing and Cloud Computing could be applied in this situation. (4 Marks). What are the main differences in how these two computing models handle resource sharing and infrastructure management? (3 Marks)

b. When developing a system that integrates multiple services from different providers, describe the key principles of Service-Oriented Architecture (SOA) that should be applied. (3 Marks). How do Loose Coupling and Reusability enhance the system's flexibility and ease of maintenance in this scenario? (4 Marks)

Composability
CRIL

c. FUTA is considering different cloud deployment models for its IT infrastructure. Explain the three main cloud deployment models that could be chosen. (3 Marks). If FUTA opts for a Public Cloud solution, describe its key characteristics and how it may benefit the institution. (3 Marks)

Public cloud
Private
Hybrid
Community

2a. Imagine FUTA is considering adopting the "Everything as a Service (XaaS)" model to streamline its operations. Describe how the XaaS model could be applied across different schools/departments and the types of services it includes. (5 Marks). Provide examples of specific XaaS solutions that could benefit the institution. (2 Marks)

System as service
Network service

b. In the context of setting up a virtualized IT environment for your institution, identify the different levels of Virtualization you might implement (3 Marks). Briefly explain each level of virtualization and how it contributes to optimizing your institution's IT infrastructure. (4 Marks)

Hardware
OS
Network

c. Evaluate the Cloud Computing model for a new project at FUTA. Highlight the benefits and drawbacks of using cloud computing, considering aspects such as cost, scalability, security, and management. (6 Marks)

3a. In the context of adopting cloud infrastructure for your institution's operations, identify the primary security concerns associated with cloud computing (4 Marks). Discuss how these security concerns can impact the overall security and integrity of data and services hosted in the cloud. (3 Marks)

b. For a project utilizing grid computing, explain the different Trust Models for Grid Security and how they ensure secure interactions between different entities (4 Marks). Describe the purpose of Grid Security Infrastructure (GSI) in maintaining grid security. (3 Marks)

Hierarchical
Public

c. When designing a grid computing system, outline the key components of the Open Grid Services Architecture (OGSA). (3 Marks). Explain the fundamental principles that underpin OGSA and how they facilitate the integration and management of grid services. (3 Marks)

4a. When considering cloud adoption for your business, discuss the various aspects of Data Security in the Cloud (4 Marks). Explain the responsibilities of both the cloud provider and the customer in ensuring data security and how these responsibilities are shared. (4 Marks)

Authorisat
Authen
Integrity
Confidentiality

b. In setting up a virtualized IT environment for FUTA, explain how Virtualization is applied across CPU, memory, and I/O devices (3 Marks). Describe how each type of virtualization enhances resource management and efficiency (3 Marks).

Hardware
OS
Network.

c. Explain the role of virtual clusters in managing resources within a cloud or grid computing environment (3 Marks). Discuss how virtual clusters contribute to optimizing resource allocation and improving system performance (3 Marks).

5a. In the context of setting up a grid computing system for a large-scale research project, explain the role of the Resource Layer in grid architecture (3 Marks). and its key functionalities (4 Marks).

b. As you implement grid computing for a collaborative project involving multiple education institutions, discuss the security challenges that may arise with grid architecture (3 Marks). How can these security challenges be effectively addressed at the Connectivity Layer? (3 Marks)

c. When virtualizing FUTA's IT infrastructure, identify the different levels of Virtualization you might use. (3 Marks). Explain how each level works. (4 Marks)

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF INFORMATION SYSTEMS

B. Tech. Degree Examination; Second Semester 2023/2024 Session

Course Title: Modelling in System Engineering; **Course Code:** IFS 508; **Credit Units:** 2

Instruction: Answer Any Three (3) Questions. Each question carries 20 Marks.

Time Allowed: 2 hours

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*1 (a) Describe four perspectives used in modelling a system? **(5 marks)**

(b) Graphical models have become most widely used approach in modelling system, why? **(5 marks)**

(c) 'FIRARS' is the name of the Information System used to manage students' academic activities and records in FUTA. As students that have used the system for your course registration and other activities for years, create a UML **Activity** for FIRARS. **(5 marks)**

(d) Briefly explain the following UML diagrams:

- i. Activity Diagrams **(1 mark)**
- ii. Use Case Diagrams **(1 mark)**
- iii. Sequence Diagrams **(1 mark)**
- iv. Class Diagrams **(1 mark)**
- v. State Diagrams **(1 mark)**

2 (a) Online learning is increasingly becoming a major learning platform for student. Create a Use-Case diagram/model that shows a student's involvement in these online learning activities. **(5 marks)**

(b) Briefly describe structural models **(5 marks)**

(c) what is the main use of Class diagrams and explain the components of a Class diagram - object attribute properties **(5 marks)**

(d) Draw a simple UML diagram showing 2 classes and association **(5 marks)**

3 (a) Generalization is a concept for modelling inheritance in OOD, explain the concept of generalization in system modelling **(5 marks)**

(b) Create a generalization model for different categories of FUTA students **(5 marks)**

(c) Describe four instruments used in Modelling as a technique LSAP **(5 marks)**

(d) Explain the four domains under which Systems Engineering can be described RBAV **(5 marks)**

Requirement
Behaviour
Architecture
Verification
Validation

*4 (a) What is system engineering? Interdisciplinary Approach **(4 marks)**

(b) Outline the objectives of a representative model - Aids decision understand. **(5 marks)**

(c) Explain benefits of Systems modelling **(5 marks)**

(d) Describe the concepts of Human factor and Human factor engineering in system modelling **(6 marks)**

CBPR.

5 (a) Explain the concept of mathematical models in system modelling **(4 marks)**

(b) create a UML diagram diagram to illustrate object aggregation in system modelling **(8 marks)**

(c) Design and Modelling are two concepts that appears confusing to engineers, clarify the concepts? **(4 marks)**

(d) Explain the concept of system thinking in systems modelling **(4 marks)**

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF INFORMATION SYSTEMS

B. Tech. Degree Examination; Second Semester 2023/2024 Session
Course Title: Information System Innovation and New Technologies; **Course Code:** IFS 502; **Credit Units: 2**
Instruction: Answer any THREE Questions ONLY. Each question carries 20 Marks.
Time Allowed: 2 hours.

Question One

a. The world has become a global village through the advent of modern technologies, discuss comprehensively the impact of globalisation on the ecosystem.(5 Marks)
b. The ability of countries to rise above narrow self-interest has brought unprecedented economic wealth and applicable scientific progress, discuss. (5 Marks)
c. There are fascinating inventions that paved way for the development of other innovations that have transformed the planet, describe four of these technological innovations. (6 Marks)
d. Propose innovative solutions to improve existing products/services in a chosen market. (4 Marks) commodi...

Question Two

a. Apple's iPhone has maintained a premium position in the smartphone market despite increasing commoditisation. Analyse the strategies Apple has employed to achieve this.(4 Marks) circl
b. Netflix has faced increased competition from streaming services like Hulu and Disney+. Analyse the variables driving substitutability in this market. .(4 Marks)
c. Analyse the diffusion process of a successful innovation, discussing the characteristics of each adopter group.(6 Marks)
d. Critically evaluate how a company leverages collective intelligence to drive innovation and performance. (6 Marks)

Question Three

a. Examine the role of web applications in enabling digital transformation for a traditional industry. (5 Marks)
b. Compare and contrast the architectural styles of APIs and web services, using examples.(6 Marks)
c. Discuss the principles of collective intelligence in accordance to theorists Don Tapscott and Anthony Williams.(4 Marks) - OPAS
d. Analyse the peer-to-peer network architecture used by BitTorrent, differentiating between structured and unstructured elements. (5 Marks)

Question Four

a. Examine the key features elements that need to be considered when designing web applications for business.(4 Marks)
b. Analyse the similarities and differences between RESTful APIs and SOAP-based web services. (5 Marks)
c. Discuss the challenges of implementing peer-to-peer networks in mobile ad-hoc networks (MANETs).(5 Marks)
d. Discuss the benefits and limitations of diffusion of Innovations theory. (6 Marks)

Question Five

a. Analyse by differentiating between centralised and distributed peer-to-peer networks.(4 Marks)
b. Discuss extensively the essence of virtual team work in an organisation and describe the various types of virtual team. (6 Marks)
c. There are a number of features that are increasingly prominent in the digital economy and which are potentially relevant from a tax perspective, describe five of them.(5 Marks)
d. Analyse the classes of social networks.with illustrative examples.(5 Marks)

TPMRS
Traditional
Discussi
media share

Network
Parallel
Production
Product Dev

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF INFORMATION SYSTEMS

B. Tech. Degree Examination; Second Semester 2023/2024 Session
Course Title: IT Security and Risk Management; **Course Code:** IFS 504;
Credit Units: 2
Instruction: Answer any THREE Questions ONLY. Each question carries 20 Marks.
Time Allowed: 2 hours.

Incident response
PDCERP

Question One

a. Cyber-attacks can be categorised based on the activities and targets of the attacker, Analyse the types of cyber-attacks, including their characteristics, goals, and potential consequences. (6 Marks)

b. Compare and contrast intrusion detection systems (IDS) and firewalls as cybersecurity solutions that can be deployed to protect an endpoint or network.(5 Marks)

c. Firewalls, intrusion detection systems, antivirus programs, and two-factor authentication products are some of the tools available to assist enterprise organisation in protecting its network and data. Write an algorithm/pseudocode for a two-factor authentication process to demonstrate this effect.(4 Marks)

d. Conduct a risk assessment using the OCTAVE (Operationally Critical Threat, Asset, and Vulnerability Evaluation) framework for a small business. Identify potential risks and recommend mitigation strategies. (5 Marks)

Question Two

a. Discuss the strengths and weaknesses of signature-based IDS solutions.(4 Marks)

b. Security breaches affect organisations in a variety of ways, discuss.(5 Marks)

c. Preventing cross-site scripting is trivial in some cases but can be much harder depending on the complexity of the application and the ways it handles user-controllable data. Discuss the effective and strong measures of preventing XSS vulnerabilities.(6 Marks) - Content security Policy parsing markup la

d. Security engineering must start early in the application deployment process, discuss the various steps that are required to achieve the process. (5 Marks)

Filter input on arrival
Analyze input
protect output.
Content security Policy

Question Three

a Assuming an organusation plans to outsource a process to an external service provider and it fails to include information security requirements in the supplier agreement due to lack of knowledge requirement, analyse the steps to set the security requirement for the supplier? (5 Marks)

b. Explain the following security measures: Rivest, Shamir, and Adleman (RSA), Advanced Encryption Standard (AES) and Data Encryption Standard (DES). (6 Marks)

c. Internet exposes computer users to risks from a wide variety of possible attacks. Write a simple algorithm for a malicious code that can detect and exploit a vulnerability in a system. (4 Marks)

d.. Discuss by differentiating between cryptography and steganography as crucial components of network security. (5 Marks)

Question Four

a. In a computer based test environment for conducting examination processes, explain the various security components that should be integrated into the environment to curb examination malpractices. (5 Marks)

b. Analyse foundational steps a company should take to establish its first customised risk assessment framework. (4 Marks) - Risk Assessment framework Start with Pinpoint and evaluate Risk. → 5 steps

c. Organisations use different access control models depending on their compliance requirements and the security levels of information systems they are trying to protect, analyse the access control models. (6 Marks) - MAC
DAC Discretionary
Role
Rule
Attribute

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
INFORMATION SYSTEMS DEPARTMENT

EXAM: B.Tech
COURSE: IFS518 System of Systems Engineering; SESSION: 2023/24, Second Semester
INSTRUCTION: Answer any 3 questions TIME: 2hrs

Question 1

a. Define system of system (SoS). The SoS can be categorized based on their degree of centrality. That is • Virtual • Collaborative • Acknowledged and • Directed. Describe them
b. To consider how a systems engineering for a system of systems different from single system, we consider **Core Elements and Crosscutting Issues**
List the Seven core elements of SoS systems engineering that provide the context for applying systems engineering processes
List **Crosscutting Issues** Note that the core SoS systems engineering elements identify systems engineering actions in an SoS, these emerging principles identify ways the elements can be implemented
c. The Defense Acquisition Guidebook defines 16 aspects of systems engineering. Describe how they apply to SoS 3DIRT 2VLC 5, 7, 8 marks

P- P- P- 1-16

Question 2.

a. Define System of Systems. The main thrust behind the desire to view the systems as an SoS
b. The largest natural SoS on our planet (i.e., the ecosystem) and all the organisms living in it are considered open systems (Kay, 2000). These systems have been evolved and sustained for millions of years by being open. Commercial and social entities are also considered open systems because they are constantly exchanging some entities. What are these entities Methods Energy Information
c. The role of the Lead System Integrator (LSI) can vary widely from acquisition to acquisition, especially in the role the LSI plays in the selection of the development contractor(s). In many cases, the LSI is responsible for integrating systems into a system of systems, but the LSI is generally not directly involved in system selection. Present any 2 definitions for the LSI 6, 7, 7 marks

Question 3

a. To survive and sustain, natural systems of systems had no choice but to adopt and follow certain fundamental open systems characteristics and principles. List them
b. What is an open system. With the aid of diagram, present the benefit and an open systems
c. Federalism is based on five principles and five axioms. These must be adopted "inside out" (Handy, 1992). A summary of Handy's five principles tailored to the domain of systems engineering and management. Highlight the five principles 5, 7, 8 marks

Question 4

a. With the aid of diagram, present the benefit of an open system 1·3AIF
b. Discuss the eight principles of system of systems engineering RDCS MESS
c. One of the principles of open system is self-government principle. True open systems are characterized by self-governance tenets. List these tenets and discuss each of the tenets

cybernetic control
Homeostasis
self organize

6, 7, 7 marks

Question 5

a. Engineering a system of systems is replete with challenges. Enumerate these challenges
b. Keating et al. (2003) make several observations relevant to engineering a system of systems and why traditional systems engineering practices are often not sufficient for engineering a system of systems. Keating et al. (2003) note several important realities concerning Traditional Versus System of System Engineering. Discuss any of the 5 realities
c. System (System of Interest). Highlight the attributes of a system of interest 5, 8, 7 marks

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
INFORMATION SYSTEMS DEPARTMENT

EXAM: B.Tech **COURSE: IFS506 Decision Support System**
SESSION: 2023/24, Second Semester
INSTRUCTION: Answer any 3 questions **TIME: 2hrs**

Question 1

a. Give two definitions of Decision support systems (DSS) Litte 1970 After 1980
b. Review the characteristics and capabilities of DSS
c. The DSS composed of the following subsystems: Data management, Model management, Communication (dialog subsystem) and the Knowledge management. Use a well annotated diagram to describe the Conceptual Model of DSS 5, 7, 8 marks

Question 2

a. DSS offer some major benefits the body of management. Highlight them
b. Discuss the data management subsystem. What are the advantages and disadvantages of the DSS database. Succinctly, discus the Database Management System (DBMS) and the various function that can be performed by the DBMS
c. Discuss User Interface (Dialog) Subsystem 6, 7, 7 marks

Question 3

a. One of the components of the DSS is the Knowledge Subsystem, in your own understanding explain the subsystem
b. Discuss the Model Management Subsystem. Emphasis should be laid on the composition of the model such as Model base, Model base management system, Modeling language, Model directory and Model execution, integration, and command
c. Succinctly, elaborate on some major capabilities of the dialog generation and management system (DGMS)

6, 7, 7 marks

Question 4

a. Decision support system have evolve simultaneously with advances in computer hardware and software technologies. List and discuss any 4 hardware options to run DSS
b. There are several ways to classify DSS. Some of the classes include the following. Discuss any 3:
Type of support: Data-oriented versus Model-oriented
Institutional versus Ad Hoc DSS
Degree of Nonprocedurality
Personal, Group, and Organizational Support
Individual versus Group DSS
Custom-made versus Ready-made systems 10, 10 marks

Question 5

a. A useful framework for understanding DSS construction issues was developed by Sprague and Carlson 1982. He identified three levels of DSS technology to include: Specific DSS, DSS generators, and DSS tools. Elaborate more on these technologies
b. Hoe does a database differ for a collection of files?
c. Models are classified as strategic, tactical, or operational. What is the purpose of such classification? Provide example of each 8, 5, 7 marks

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

Information systems Department

EXAM: B.Tech **COURSE: CSC Special Topics IFS512**

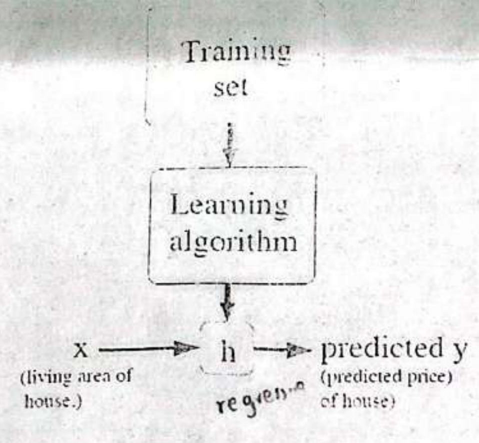
INSTRUCTION: Answer question 1 and any 2 other questions **Time: 1.5hr**

Question One

a. Read through the following statements and fill in the gaps labelled from a to i

To establish notation for future use, we'll use $x^{(i)}$ to denote the "input" variables (living area in this example), also called input feature ____a____ and $y^{(i)}$ to denote the ____________b____ variable that we are trying to predict (price). A pair $(x^{(i)}, y^{(i)})$ is called a __________c________ and the dataset that we'll be using to learn—a list of m training examples $\{(x^{(i)}, y^{(i)}); i = 1, \ldots, m\}$—is called a ________d___. Note that the superscript "(i)" in the notation is simply an index into the training set, and has nothing to do with exponentiation. We will also use $\mathcal{X}$ denote the space of ______e_____ and $\mathcal{Y}$ the space of ________f___. In this example, $\mathcal{X} = \mathcal{Y} = \mathbb{R}$.

To describe the supervised learning problem slightly more formally, our goal is, given a training set, to learn a function $h : \mathcal{X} \mapsto \mathcal{Y}$ so that $h(x)$ is a "good" predictor for the corresponding value of y. For historical reasons, this function h is called a ________g_. Seen pictorially, the process is therefore like this:



When the target variable that we're trying to predict is continuous, such as in our housing example, we call the learning problem a regression ____h____ problem. When y can take on only a small number of discrete values (such as if, given the living area, we wanted to predict if a dwelling is a house or an apartment, say), we call it a classification ____i____ problem.

2 marks each

Question Two

a. Machine learning is the field of study that gives computers the ability to **learn** without being explicitly programmed. Define "Learning" according to Tom Mitchell's description from his Machine Learning book using terms such as; experience E, class of tasks T and performance measure P

b. The three broad categories of machine learning are summarized in the following: Supervised learning, unsupervised learning, and reinforcement learning. Explain each and give an example for each

c. A neural network is basically a realization of a nonlinear mapping from $\mathbb{R}^I$ *to* $\mathbb{R}^K$

$$\mathcal{F}_{NN}: \mathbb{R}^I \rightarrow \mathbb{R}^K$$

where I and K are respectively the dimension of the input and target (desired output) space. The function $\mathcal{F}_{NN}$ is usually a complex function of a set of nonlinear functions, one for each neuron in the network. Neurons form the basic building blocks of NNs. Discuss the single neuron, that is also referred to as the *perceptron*, in detail. 6, 7, 7 marks

Question Three

a. List 6 Classes of Machine Learning Algorithms
b. There are several different components of machine learning algorithms. Discuss the most important five
c. To aid our conceptual understanding, each of the machine learning algorithms can be categorized into various categories as eager vs lazy; batch vs online; parametric vs nonparametric; discriminative vs generative. Discuss each one 5, 7, 8 marks

Question Four

a. As indicated in the figure of Pedro Domingo's 5 Tribes of Machine Learning, there are several different components of machine learning algorithms. Itemize and describe them
b. In Least Mean Squares (LMS) algorithm, we want to choose θ so as to minimize $J(\theta)$. To do so, let's use a search algorithm that starts with some "initial guess" for θ, and that repeatedly changes θ to make $J(\theta)$ smaller, until hopefully we converge to a value of θ that minimizes $J(\theta)$. Specifically, let's consider the gradient descent algorithm, which starts with some initial θ, and

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

repeatedly performs the update:
(This update is simultaneously performed for all values of $j = 0, \ldots, n.$) Here, α is called the learning rate. This is a very natural algorithm that repeatedly takes a step in the direction of steepest decrease of J. In order to implement this algorithm, we have to work out what is the partial derivative term on the right hand side of the equation.
Question: work it out for the case of if we have only one training example (x, y), so that we can neglect the sum in the definition of J. Use the above illustration to define the rule called the LMS update rule and is also known as the Widrow-Hoff learning rule. Find in closed-form the value of θ that minimizes $J(\theta)$. i.e. $\theta = (X^T X)^{-1} X^T \vec{y}$. 10, 10 marks

✔**Question Five**

a. a generic biometric system can be viewed as having four main modules: a sensor module; a quality assessment and feature extraction module; a matching module; and a database module. For each of these modules described
b. A number of biometric characteristics are being used in various applications. Each biometric has its pros and cons and, therefore, the choice of a biometric trait for a particular application depends on a variety of issues besides its matching performance. Jain et al. [12] have identified seven factors that determine the suitability of a physical or a behavioral trait to be used in a biometric application. Discuss these factors - 4 4
c. Present a brief introduction to some of the commonly used biometric characteristics in this modern society \ iris 7, 6, 7 marks

Fingerprint
Hand geometry
Face geometry

Universal
Uniqueness
Performance
Permanence
Convenience
Availability.

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF INFORMATION SYSTEMS
B. Tech. Degree Examination; Second Semester 2023/2024 Session
Course Title: Performance Evaluation and Benchmarking;
Code: IFS 514; **Units:** 2
Instruction: Answer *three* Questions in ALL. **Time:** 2hrs

Question 1

Response time
throughput

a. Explain some of the commonly used performance metrics? **(10 marks)**

b. The Systems Performance Evaluation Cooperative (SPEC) is a nonprofit corporation formed by leading computer vendors to develop a standardized set of benchmarks in Release 1.0 of the SPEC benchmark suite. Highlight the 10 benchmarks drawn from various engineering and scientific applications **(10 marks)**

- Response time
Processor speed
Performance ~~sp~~ speed
Reliability
Memory space
Buffer size

Question 2

a. Discuss two methods for forecasting demand in an organization and explain how each method can improve decision-making. **(5 Marks)**

b. Explain the concept of capacity requirements planning and discuss how an organization can address capacity gaps. **(5 Marks)**

c. A parallel processing system completes a task in 110 seconds, while the same task takes 1100 seconds on a single processor. Calculate the speedup achieved by using the parallel processing system. **(5 Marks)**

d. Two servers are benchmarked for energy efficiency. Server X completes a task using 5000 Joules, and Server Y completes the same task using 200 Joules. Calculate the energy efficiency index for Server Y compared to Server X. **(5 Marks)**

Question 3

Students arrive at computer lab in 10 per hour. Spend 20 minutes at a terminal (assume exponential distribution) and then leave. Assume the Centre has 5 terminals.

i. How many terminals can go down and still be able to service the students? (5 marks)

ii. What is the probability that all terminals are busy? (5 marks)

iii. How long will a student wait in the Centre? (5 marks)

iv. What would happen if the terminals were distributed in separate labs, one per lab across campus? Comment on the waiting time in this scenario (5 marks)

Question 4

a. Discuss the following techniques of software monitoring (i) Trap Instruction (ii) Trace Mode (iii) Timer Interrupt (6 Marks)

b. Discuss some important reasons for monitoring program execution (4 Marks)

c. What is an accounting log? Enumerate the typical data that is contained in an accounting log (10 Marks)

Question 5

a. Differentiate between clustering and classification. Explain the situations in which they can be used (10 marks)

b. Highlight 10 common mistakes in benchmarking (10 marks)